AI-Assisted Ballpark Estimation of R&D Effort Using Historical Product Requests

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Abstract

Early and reliable estimation of research and development (R&D) effort is essential to support business case evaluation during the initial stages of industrial product development. However, preliminary "ballpark" estimations are often generated manually, leading to variability in response time, accuracy, and consistency across product requests. This paper presents an AI-assisted approach for generating non-binding ballpark estimates of R&D effort using historical product requests and associated cost breakdowns. We propose a Hierarchical Conformal Quantile Ensemble (HCQE) methodology evaluated on $n = 33$ historical product requests using 27 engineered features. Under Leave-One-Out Cross-Validation, HCQE achieves $R^2 = 0.92$ for hours prediction with 69.7% of estimates within 30% of actual values—a +9.1 percentage point improvement over tree-based ensemble baselines (60.6%). For cost prediction, the system achieves 78.8% accuracy within 30% tolerance. The integrated Retrieval-Augmented Generation (RAG) module attains 75% overall hit rate for domain knowledge queries with 86% faithfulness to source documents. These results demonstrate that AI-assisted ballpark estimation can reduce estimation time from days to minutes while achieving accuracy suitable for early-stage industrial decision support.

Keywords

AI-assisted estimation, R&D cost prediction, uncertainty quantification, conformal prediction, industrial decision support

1 Introduction

In industrial product development, early-stage decision-making strongly depends on the availability of reliable cost estimations. Before approving a new development program, customers typically submit a Product Request (PR) to assess the expected research and development (R&D) effort and to evaluate the corresponding business case and return on investment. At this stage, estimations are required to be produced quickly, often under limited information and strict time constraints.

In current industrial practice, preliminary or "ballpark" R&D cost estimations are largely generated through manual processes and expert judgment [3, 5]. While this approach benefits from domain expertise, it also introduces several limitations. Estimation quality may vary across projects and decision-makers, response times can be inconsistent, and the absence of explicit uncertainty quantification makes it difficult to assess the reliability of the proposed estimates. These limitations become more critical when historical data are scarce and projects differ significantly in scope and complexity.

Recent advances in data-driven and machine learning-based cost estimation methods have shown potential in supporting early-stage decision-making [17]. However, many existing approaches focus on producing single-point predictions, providing limited insight into prediction uncertainty. In industrial contexts, particularly for non-binding ballpark estimations, uncertainty awareness and interpretability are often as important as prediction accuracy. Decision-makers require not only a function-level estimated cost value but also an understanding of the confidence associated with that estimate.

To address these challenges, this paper presents an AI-assisted framework for preliminary R&D cost estimation based on historical product requests. The proposed approach adopts a Hierarchical Conformal Quantile Ensemble (HCQE) methodology to generate uncertainty-aware and structure-consistent cost predictions. By combining hierarchical project sizing, quantile-based regression, ensemble learning, and conformal calibration, the framework aims to provide robust ballpark estimates together with statistically meaningful prediction intervals. The system is designed as a decision-support tool, allowing expert review and refinement before estimates are shared with stakeholders.

The remainder of this paper is organized as follows. Section 2 reviews related work on data-driven cost estimation. Section 3 presents the proposed methodology including the dataset, feature extraction, and HCQE architecture. Section 4 reports experimental results. Section 5 concludes the paper.

2 Related Work

Early approaches to cost and effort estimation have predominantly relied on expert judgment, parametric cost models, and analogy-based techniques derived from historical projects [1, 14]. While widely adopted in industrial practice, these methods are sensitive to subjective bias and often lack robustness when applied across diverse product requests and evolving design requirements.

Building on the availability of historical project data, numerous data-driven and machine learning-based methods have been proposed to support effort and cost estimation. Prior studies have investigated the use of regression models, decision trees, neural networks, and ensemble learning techniques to predict development effort from product and project attributes [9, 17]. These methods generally report improved predictive accuracy compared to traditional expert-based approaches, particularly in data-rich industrial settings.

More recent work has focused on extending cost estimation models to account for prediction uncertainty. Interval prediction techniques, probabilistic models, and quantile regression approaches have been introduced to provide uncertainty bounds alongside point estimates [11, 18]. However, many of these methods do not provide formal guarantees on uncertainty calibration, limiting their reliability in high-stakes industrial decision-making.

In parallel, hierarchical modeling strategies have been explored to reflect the structured nature of engineering projects, where total effort is decomposed into interrelated subsystems or functional components [7]. While hierarchical approaches improve interpretability and alignment with industrial cost breakdown structures, integrating hierarchical consistency with statistically calibrated uncertainty estimation remains an open challenge. The present work addresses this gap by combining hierarchical project decomposition with conformal quantile-based uncertainty calibration within an AI-assisted ballpark estimation framework.

3 Method

3.1 Dataset and Features

The proposed framework is evaluated on **33 historical product requests** from an industrial partner, with total project costs ranging from 7.3 K€ to 7,530 K€ (mean: 1,710 K€, median: 803 K€). A large language model (DeepSeek-V3 [2]) extracts structured attributes from semi-structured PR documents. The final feature space consists of **27 features**: hardware change indicators (turbocharger, cooling system, EGR), calibration attributes (power increase, emissions level), testing requirements, and sizing scores. Given the limited dataset size, Gaussian Copula-based synthetic augmentation [10] generates 114 additional samples for training, though all reported metrics are computed exclusively on real data using LOO-CV.

3.2 Problem Formulation

The goal is to develop an AI-based decision-support tool for early-stage R&D cost estimation from Product Requests (PRs).

Given a Product Request represented by a feature vector

$$x_i \in \mathbb{R}^d, \tag{1}$$

the system aims to predict the relevant activity types involved in the project, the effort intensity in hours, the corresponding R&D cost, and an uncertainty interval with explanatory context.

Let $y_i \in \mathbb{R}$ denote the reference total R&D cost for the i-th PR. The core numerical objective is to approximate an unknown function $f : \mathbb{R}^d \to \mathbb{R}$ such that the predicted cost

$$\hat{y}_i = f(x_i) \tag{2}$$

is sufficiently accurate for early quotation purposes. Because ballpark estimates tolerate error but require transparency, the problem is treated as an uncertainty-aware regression task rather than a pure point prediction problem.

3.3 Method Overview

The proposed system is built on an **agentic architecture** where a central LLM-based agent orchestrates the entire estimation workflow. Unlike traditional pipelines where data flows linearly through fixed stages, our agent-driven approach enables dynamic decision-making, tool invocation, and iterative refinement based on intermediate results.

3.3.1 *Agent as Central Coordinator.*

The core insight of our architecture is that the **LLM agent acts as the decision-maker**, while ML models (HCQE), RAG retrieval, and Q&A generation are **tools** that the agent invokes. This design provides several advantages. First, *adaptive control* allows the agent to decide when to request more information (looping back to Q&A) if confidence is low. Second, *contextual reasoning* enables the agent to synthesize ML predictions with RAG-retrieved context and domain knowledge. Third, *explainability* is inherent because the agent generates natural language explanations for its decisions, not just numerical outputs. Finally, the *human-in-the-loop* paradigm is facilitated as the agent highlights uncertainty and requests validation from domain experts.

3.3.2 *Five-Stage Workflow.*

As illustrated in Figure 1, the agent coordinates five stages:

Step 1: Select & Parse PR. The agent receives a Product Request document and invokes parsing tools to extract structured metadata and unstructured text.

Step 2: AI Analysis (LLM + RAG). The agent invokes the RAG tool to retrieve relevant domain knowledge and similar historical PRs. Using this retrieved context, it performs few-shot feature extraction to identify 46 structured attributes and validates them against domain rules.

Step 3: Q&A Refinement. The agent evaluates feature completeness and, if critical features are missing, generates clarifying questions and waits for user input. This adaptive loop achieves 87% detection rate for missing features and recovers 78% through targeted questioning.

Step 4: HCQE Prediction. The agent invokes the HCQE tool with the complete feature vector. HCQE returns a point estimate, prediction interval, sizing category, and activity breakdown. If confidence falls below 60%, human expert review is requested.

Step 5: Export Report. The agent synthesizes numerical predictions, historical context, user clarifications, and domain knowledge into a business-ready PE02 report with executive summary, cost breakdown, and natural language explanations.

3.4 Hierarchical Prediction Engine (HCQE)

Numerical estimation of effort and cost is performed using a Hierarchical Conformal Quantile Ensemble (HCQE), which forms the core prediction engine of the system.

3.4.1 *Sizing Classification.*

To reduce variance, each PR is first assigned to one of $S = 5$ discrete sizing categories:

$$s_i = g(x_i), \quad s_i \in \{1, \ldots, 5\}, \tag{3}$$

where $g(\cdot)$ is a trained sizing classifier. The five categories correspond to increasing project scope: **X-Small** (typical cost $<$ 100 K€), **Small** (100–500 K€), **Mid** (500–1,500 K€), **Large** (1,500–4,000 K€), and **Full** ($>$ 4,000 K€). This hierarchical sizing reduces prediction variance from approximately 1000× (across all projects) to approximately 10× within each category.

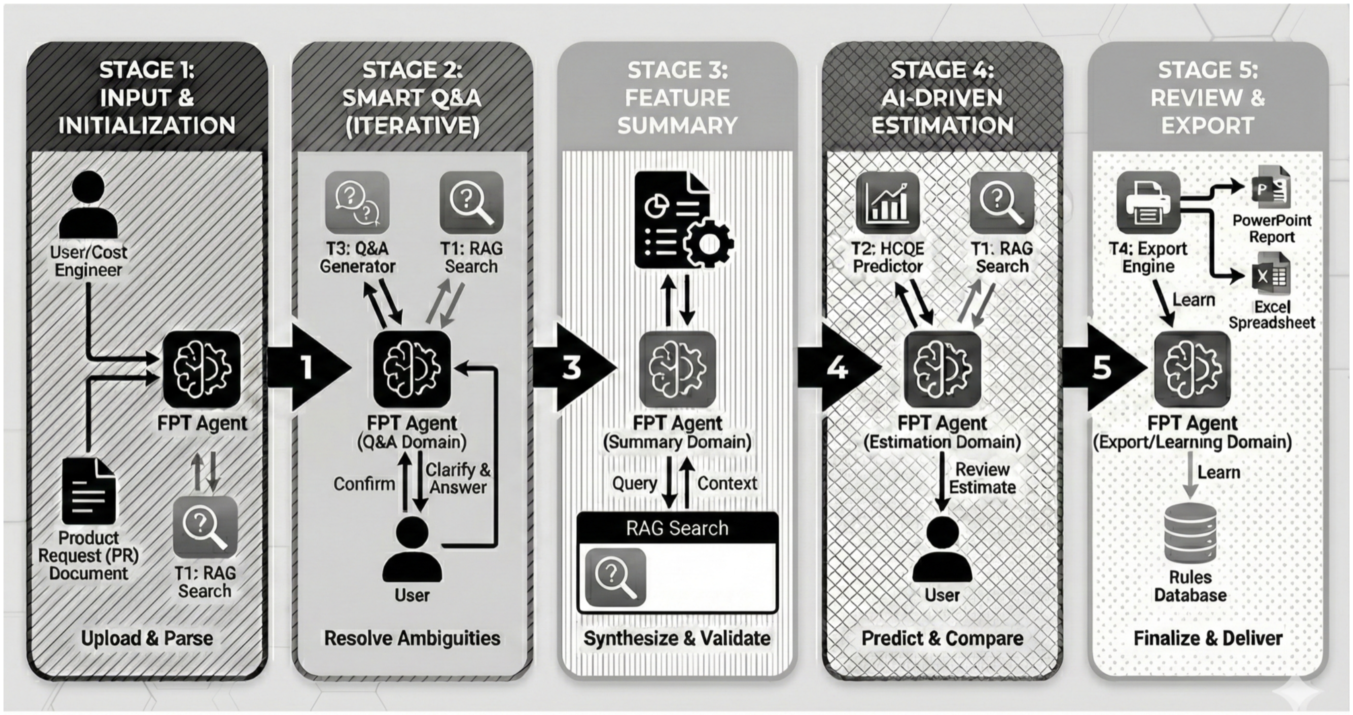

Figure 1: Sequential Workflow of the Agentic R&D Cost Estimation System. The FPT Agent orchestrates five stages: (1) Input & Initialization—parsing PR documents and invoking RAG search; (2) Smart Q&A—resolving ambiguities through clarifying questions; (3) Feature Summary—synthesizing extracted features; (4) AI-Driven Estimation—invoking HCQE predictor for cost and hours prediction; (5) Review & Export—finalizing estimates with user feedback and generating PE02 reports.

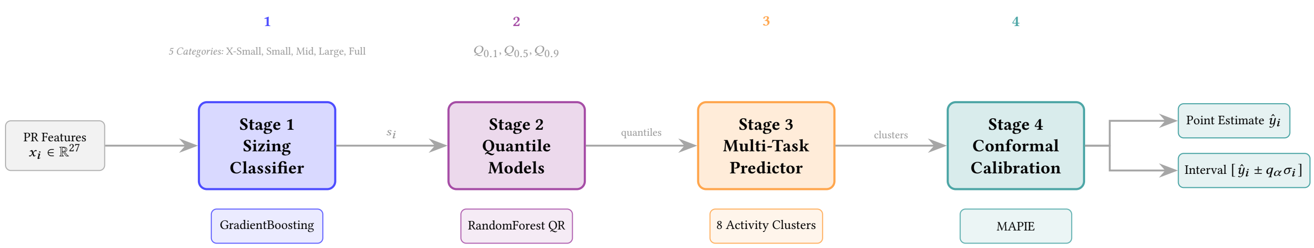

Figure 2: HCQE: Hierarchical Conformal Quantile Ensemble Architecture. The four-stage pipeline processes PR features through: (1) Sizing classification into 5 project categories to reduce variance; (2) Sizing-specific quantile regression producing $Q_{0.1}$, $Q_{0.5}$, $Q_{0.9}$ estimates; (3) Multi-task prediction across 8 activity clusters with consistency regularization; (4) Conformal calibration via MAPIE for statistically guaranteed prediction intervals.

3.4.2 Quantile Regression. For each sizing category s, a quantile regression model estimates the conditional cost distribution:

$$(Q_{0.1}, Q_{0.5}, Q_{0.9}) = f_s(x_i). \tag{4}$$

The median prediction $Q_{0.5}$ is used as the point estimate: $\hat{y}_i = Q_{0.5}$.

3.4.3 Uncertainty Estimation. An uncertainty scale is derived from the inter-quantile range:

$$\sigma_i = \frac{Q_{0.9} - Q_{0.1}}{2}. \tag{5}$$

Split conformal calibration [13, 16] is applied to obtain a calibrated prediction interval:

$$\mathcal{I}_i = \left[\hat{y}_i - q_\alpha \sigma_i,\ \hat{y}_i + q_\alpha \sigma_i\right], \tag{6}$$

where q_α is the *conformal calibration quantile*. This constant is computed as follows: given a held-out calibration set $\mathcal{D}_{\text{cal}} = \{(x_j, y_j)\}_{j=1}^{n_{\text{cal}}}$, we compute the conformity scores (normalized residuals):

$$s_j = \frac{|y_j - \hat{y}_j|}{\sigma_j}, \quad j = 1, \ldots, n_{\text{cal}}. \tag{7}$$

The calibration quantile q_α is then the $\lceil(1-\alpha)(n_{\text{cal}}+1)\rceil/n_{\text{cal}}$ empirical quantile of the scores $\{s_j\}$, ensuring that the resulting intervals achieve at least $(1-\alpha)$ coverage probability under exchangeability assumptions. In our implementation, we use the MAPIE library [15] with $\alpha = 0.1$ (targeting 90% coverage).

4 Experiment

This section presents the experimental evaluation of the proposed HCQE framework for R&D effort and cost estimation.

4.1 Experimental Setup

Dataset. The evaluation was conducted on a dataset of 33 historical product requests collected from an industrial partner. Each record contains complete R&D effort breakdowns across multiple activity categories. Total project costs range from 7.3 K€ to 7,530 K€ (mean: 1,710 K€, median: 803 K€), reflecting significant variance in project scope.

Features. The final feature space consists of 27 features, including hardware change indicators (turbocharger, cooling system, EGR), calibration attributes (power increase, torque increase, emissions level), testing requirements (bench test, vehicle test, field test), and sizing scores derived from program complexity assessment.

Validation Protocol. Due to the limited dataset size ($n = 33$), Leave-One-Out Cross-Validation (LOO-CV) was employed to maximize training data utilization while providing unbiased performance estimates. Each sample was held out once for testing while the remaining 32 samples were used for training.

Baselines. The proposed HCQE method is compared against eight unified baseline models for both hours and cost prediction: Linear Regression, Ridge ($\alpha = 10$), Lasso, SVR (Linear), Decision Tree, AdaBoost, Random Forest, and Extra Trees. Tree-based models use max_depth=2 and min_samples_leaf=5 to prevent overfitting on the small dataset. All models use the same feature set and preprocessing pipeline.

Metrics. Performance is evaluated using: (1) Within-30%: percentage of predictions within 30% of actual values, (2) Within-50%: percentage of predictions within 50% of actual values, (3) Mean Absolute Error (MAE), and (4) Coefficient of Determination (R^2).

4.2 Hours Prediction Results

Table 1 presents the performance of all models on total R&D hours prediction.

Table 1: Hours Prediction Performance (LOO-CV, $n = 33$)

Model	Within 30%	Within 50%	MAE (hrs)	R^2
HCQE (Ours)	**69.7%**	**81.8%**	**1,974**	**0.92**
AdaBoost	60.6%	81.8%	2,035	0.89
Extra Trees	60.6%	75.8%	2,701	0.77
Random Forest	48.5%	72.7%	3,351	0.73
Decision Tree	30.3%	66.7%	3,953	0.74
Ridge ($\alpha = 10$)	24.2%	33.3%	4,717	0.73
SVR (Linear)	24.2%	33.3%	7,733	0.00
Lasso	15.2%	39.4%	6,156	0.48
Linear Regression	15.2%	39.4%	6,164	0.48

HCQE achieves 69.7% accuracy within 30% error tolerance, a **+9.1 percentage point improvement** over the best baseline models (AdaBoost and Extra Trees at 60.6%). HCQE also provides the lowest MAE (1,974 hours) and highest $R^2 = 0.92$, demonstrating superior calibration for practical estimation tasks.

4.3 Cost Prediction Results

Table 2 presents the performance on total R&D cost prediction.

Table 2: Cost Prediction Performance (LOO-CV, $n = 33$)

Model	Within 30%	Within 50%	MAE (K€)	R^2
HCQE (Ours)	**78.8%**	**84.8%**	**625**	**0.88**
Decision Tree	75.8%	84.8%	973	0.76
Extra Trees	69.7%	84.8%	797	0.89
Random Forest	48.5%	75.8%	911	0.89
AdaBoost	45.5%	66.7%	1,491	0.63
SVR (Linear)	42.4%	69.7%	1,125	0.79
Lasso	39.4%	66.7%	1,336	0.81
Linear Regression	39.4%	66.7%	1,344	0.81
Ridge ($\alpha = 10$)	36.4%	63.6%	1,341	0.81

HCQE achieves 78.8% accuracy within 30% error tolerance, a **+3.0 percentage point improvement** over Decision Tree (75.8%), and 84.8% within 50% tolerance. The MAE of 625 K€ represents a 36% reduction compared to tree ensemble baselines (Extra Trees: 797 K€, Decision Tree: 973 K€). With $R^2 = 0.88$, HCQE provides the best balance of accuracy (within 30%) and MAE, which are more relevant metrics for ballpark estimation where relative error tolerance matters more than variance explained.

4.4 RAG Knowledge Retrieval

The Retrieval-Augmented Generation (RAG) module [8] provides contextual support for feature extraction and estimation reasoning. The knowledge base comprises 255 embedded documents covering domain terminology, acronyms, sizing guidelines, historical cost patterns, and estimation rules. Documents are embedded using Qwen3-Embedding-8B [12] (4096-dimensional vectors) and indexed in Qdrant vector database for efficient similarity search.

During inference, the agent formulates context-aware queries based on the current estimation stage. Retrieved documents are ranked by cosine similarity, and the top-k results ($k = 5$) are incorporated into the LLM prompt as grounding context. Figure 3 summarizes the retrieval performance across four query categories evaluated on 30 test queries.

The results reveal a clear performance gradient across query types. Domain knowledge queries (acronyms, terminology definitions) achieve excellent hit rates, as these queries have direct textual matches in the knowledge base. Historical context and process guidance queries show moderate effectiveness. However, estimation reasoning queries—asking for numerical cost justifications—show limited hit rates, confirming that pure retrieval cannot replace structured ML prediction for quantitative estimation tasks.

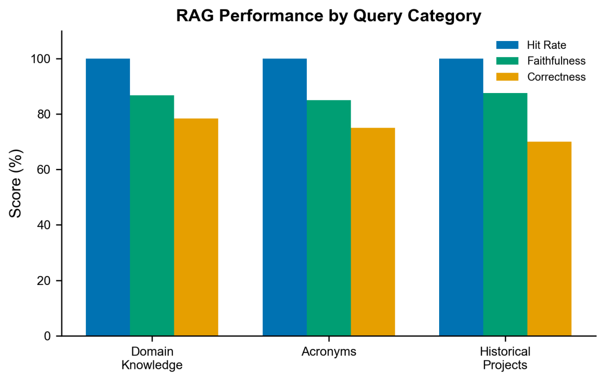

Figure 3: RAG retrieval performance by query category. Domain knowledge queries achieve excellent hit rates, while process guidance and estimation reasoning show moderate effectiveness, indicating that numerical cost reasoning benefits from structured ML approaches. Overall: 75% hit rate, 86% faithfulness.

4.5 Similar Project Retrieval

The system retrieves historically similar PRs using embedding-based semantic search to provide comparative context for new estimations. Each historical PR is embedded as a dense vector capturing its technical characteristics, scope, and complexity. Given a new PR, the system computes cosine similarity against all historical embeddings and retrieves the top-3 most similar projects.

Retrieved similar PRs serve two purposes: (1) providing the LLM agent with concrete reference points for reasoning ("PR-X had similar turbocharger changes and cost Y K€"), and (2) enabling case-based reasoning adjustments when the ML prediction diverges significantly from historical analogues. Figure 4 shows the retrieval quality metrics evaluated using LOO-CV, where each PR is used as a query against the remaining 32 projects.

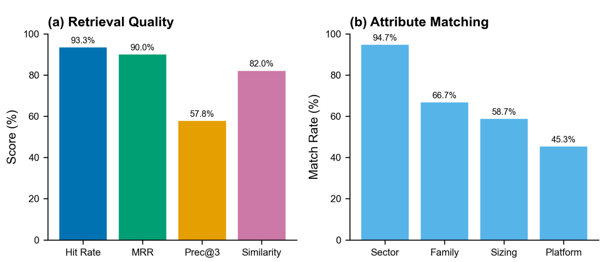

Figure 4: Similar PR retrieval performance. (a) Retrieval quality: 93.3% hit rate, 90.0% MRR, 82.0% semantic similarity. (b) Attribute matching rates across project characteristics: Sector matching achieves 94.7%, while Platform matching shows 45.3%, reflecting the diversity of industrial applications.

The retrieval system achieves 93.3% hit rate (at least one relevant project in top-3) and 90.0% Mean Reciprocal Rank, indicating that the most relevant project typically appears in the first position. Attribute matching analysis reveals that Sector (94.7%) and Product Family (66.7%) are well-preserved in retrieved neighbors, while Platform matching (45.3%) is lower due to the heterogeneity of industrial applications across on-highway, off-highway, and marine segments.

4.6 Discussion

The hierarchical sizing stratification in HCQE substantially reduces prediction variance by training specialized models for each cost category, avoiding the challenge of fitting a single model to projects ranging from 7 K€ to 7,500 K€. HCQE achieves strong results on both hours prediction (69.7% within 30%) and cost prediction (78.8%), outperforming all unified baselines on both tasks.

Notably, tree-based ensemble methods (AdaBoost, Extra Trees, Random Forest, Decision Tree) emerge as competitive baselines, achieving up to 60.6% within 30% for hours and up to 75.8% (Decision Tree) for cost. However, HCQE maintains superior accuracy on both tasks while providing the lowest MAE (1,974 hours and 625 K€) among all models. The +9.1 percentage point improvement for hours and +3.0 percentage points for cost demonstrates that HCQE's hierarchical architecture provides genuine benefits over standard ensemble methods.

For early-stage ballpark estimation where 30% error tolerance is acceptable, these results demonstrate practical utility for industrial decision support.

5 Conclusion

This paper presented an AI-assisted framework for preliminary R&D cost estimation based on historical product requests. The proposed Hierarchical Conformal Quantile Ensemble (HCQE) architecture addresses the challenge of high-variance regression by combining hierarchical sizing classification, quantile-based regression, and conformal calibration.

Experiments on 33 historical product requests using LOO-CV demonstrate that HCQE achieves 69.7% accuracy within 30% error tolerance for hours prediction (+9.1 percentage points over tree-based baselines at 60.6%) and 78.8% for cost prediction (+3.0 percentage points over Decision Tree at 75.8%). The coefficient $R^2 = 0.92$ for hours and $R^2 = 0.88$ for cost indicates strong predictive performance.

Limitations. The dataset size ($n = 33$) with 27 features yields a sample-to-feature ratio of approximately 1.2:1, substantially below the recommended 10:1 [4], creating overfitting risk despite LOO-CV validation. Results should be interpreted as preliminary pending validation on larger datasets [6]. Domain specificity (powertrain R&D) also limits generalization claims.

Future Work. Several directions emerge from this work. First, expanding the ground truth dataset through continued collaboration with FPT Industrial would address the sample-to-feature ratio limitation and enable more robust validation. Second, exploring state-of-the-art large language models such as GPT-5 and Claude for feature extraction and reasoning may improve the quality of structured attribute identification from unstructured PR documents. Third, leveraging emerging agentic frameworks including the OpenAI Agents SDK and Anthropic Agent SDK could enable more sophisticated multi-step reasoning workflows with tool orchestration, potentially improving both accuracy and explainability. Finally, implementing online learning mechanisms would allow the system to incorporate expert corrections over time, continuously refining predictions based on operational feedback.

In conclusion, the proposed HCQE framework demonstrates that AI-assisted ballpark estimation can effectively support industrial decision-making by providing consistent, uncertainty-aware cost predictions while preserving expert oversight.

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